

## Project Executable Files & Deployment Specification

|                      |                 |                       |                  |
|----------------------|-----------------|-----------------------|------------------|
| <b>Date:</b>         | 3 October 2026  | <b>Team ID:</b>       | unm1611MADHANRAJ |
| <b>Project Name:</b> | AI FitTrack API | <b>Maximum Marks:</b> | 3 Marks          |

### Submission Checklist:

This document verifies that all executable and deployable deliverables for AI FitTrack API are provided and documented for evaluator execution and grading.

| S.No | Item to Submit                                       | Submitted | File / Location Reference                                           |
|------|------------------------------------------------------|-----------|---------------------------------------------------------------------|
| 1    | Complete source code (all files and folders)         | Yes       | Code Files/src/ (controllers, models, routes, services, middleware) |
| 2    | README / Setup Guide (execution instructions)        | Yes       | Code Files/README.md                                                |
| 3    | Dependency manifest (package.json / lock)            | Yes       | Code Files/package.json & package-lock.json                         |
| 4    | Database schema & ODM configuration                  | Yes       | src/models/User.js, src/models/Workout.js, src/config/db.js         |
| 5    | Environment configuration file (.env.example / .env) | Yes       | Code Files/.env (PORT, MONGO_URI, JWT_SECRET, GEMINI_API_KEY)       |
| 6    | Deployed application / Localhost server URL          | Yes       | http://localhost:5000 / Cloud Production Ready                      |
| 7    | API Testing Suite / Postman collection               | Yes       | Code Files/FitTrack.postman_collection.json                         |
| 8    | Environment variables for testing client             | Yes       | Postman collection variables (jwt_token, workout_id)                |
| 9    | Test files and test result verification              | Yes       | Performance Testing.pdf & Postman execution logs                    |
| 10   | Project Demonstration Video walkthrough              | Yes       | FitTrack API Demo.mp4 (Full feature video walkthrough)              |

## Project Executable Files — File & Folder Structure (Part 1)

|                      |                 |                       |                  |
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### Step 2: File / Folder Structure Directory Tree:

#### AI-FITTRACK-API REPOSITORY STRUCTURE

```
FitTrack/
├── src/
│   ├── config/
│   │   └── db.js # MongoDB Atlas connection handler via Mongoose
│   ├── controllers/
│   │   ├── authController.js # User registration, login & profile controller
│   │   ├── workoutController.js # Full workout CRUD & regex search logic
│   │   └── aiController.js # Gemini workout plan & fitness insights controller
│   ├── middleware/
│   │   ├── auth.js # JWT Bearer token authentication guard
│   │   └── errorHandler.js # Centralized Express error interceptor
│   ├── models/
│   │   ├── User.js # Mongoose User schema (name, email, password)
│   │   └── Workout.js # Mongoose Workout schema (name, category, duration, calories)
│   ├── routes/
│   │   ├── authRoutes.js, workoutRoutes.js, aiRoutes.js, index.js
│   └── services/
│       ├── geminiService.js # Google Gemini AI SDK (@google/genai) service wrapper
│       ├── app.js, server.js # Express application setup and server entry point
│       ├── .env, .env.example, .gitignore, package.json, package-lock.json
│       └── FitTrack.postman_collection.json # Comprehensive automated Postman test suite
```

### Core Source Files Breakdown:

| File Path                            | Size / Metric | Primary Functionality                                                                           |
|--------------------------------------|---------------|-------------------------------------------------------------------------------------------------|
| src/server.js                        | 25 lines      | Loads dotenv, connects to MongoDB via connectDB(), boots Express HTTP server.                   |
| src/app.js                           | 35 lines      | Mounts CORS, JSON body parser, Morgan logger, routes/index.js, and error middleware.            |
| src/controllers/authController.js    | 65 lines      | Implements register (bcrypt hash), login (password compare & JWT generation), and profile.      |
| src/controllers/workoutController.js | 95 lines      | Implements createWorkout, getAllWorkouts, getWorkoutById, updateWorkout, deleteWorkout, search. |
| src/controllers/aiController.js      | 55 lines      | Implements generateWorkoutRecommendation and generateFitnessInsights with Gemini AI.            |
| src/services/geminiService.js        | 50 lines      | Initializes @google/genai SDK, constructs fitness prompts, handles AI response streams.         |

## Project Executable Files — Deployment & Environment Details

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### Step 3: Deployment & Access Details:

- **Local Server URL:** <http://localhost:5000> (Local Development Environment)
- **Production Cloud URL:** <https://fittrack-api.onrender.com> (Render Cloud Production)
- **Demo Login Credentials:** Email: john@gmail.com | Password: 123456 (Pre-seeded demo user)
- **Hosting Infrastructure:** Node.js Runtime Environment / Render Web Service / MongoDB Atlas Cloud Cluster
- **Repository Reference:** Local workspace deliverables / Git Version Controlled
- **Demo Video Walkthrough:** FitTrack API Demo.mp4 (High-resolution demonstration video)

### Environment Configuration Specifications (.env):

| Variable Key   | Example Value / Format     | Security Role & Purpose                                                              |
|----------------|----------------------------|--------------------------------------------------------------------------------------|
| PORT           | 5000                       | Defines the local TCP network port where Express listens for incoming HTTP requests. |
| MONGO_URI      | mongodb+srv://.../fittrack | Encrypted TLS connection string to MongoDB Atlas cloud database cluster.             |
| JWT_SECRET     | your_jwt_secret_key_here   | Private cryptographic secret key used to sign and verify HMAC-SHA256 JWT tokens.     |
| GEMINI_API_KEY | AlzaSy... (64-char key)    | Google Cloud AI Studio secret key authorizing generative LLM inference requests.     |

## Project Executable Files — Run Instructions & Known Issues

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### Step 4: Step-by-Step Local Run Instructions:

**Step 1 — Prerequisites:** Ensure Node.js (version 20 or higher) and npm are installed. Verify with `node -v`.

**Step 2 — Navigate to Project Directory:** Open terminal in the codebase folder: `cd "Code Files"`.

**Step 3 — Install Dependencies:** Execute `npm install` to download all packages specified in `package.json`.

**Step 4 — Configure Environment:** Create a `.env` file based on `.env.example` with valid `MONGO_URI` and `GEMINI_API_KEY`.

**Step 5 — Start Server:** Execute `npm run dev` (development reload) or `npm start` (production mode). Terminal displays: *Server running on port 5000 and MongoDB Connected.*

**Step 6 — Postman Test Execution:** Import `FitTrack.postman_collection.json` in Postman and run the entire collection. All 10 requests pass with 200/201 status codes!

### Step 5: Known Issues / Limitations & Workarounds:

| S.No | Known Limitation / Issue              | Workaround / Mitigation Strategy                                                                         | Status    |
|------|---------------------------------------|----------------------------------------------------------------------------------------------------------|-----------|
| 1    | Gemini Free Tier Rate Limits (15 RPM) | Optimized concise prompt tokens and implemented graceful fallback handling.                              | Mitigated |
| 2    | Cold Start on Free MongoDB Cluster    | Connection pooling established at server boot ( <code>server.js</code> ) with keep-alive socket options. | Resolved  |
| 3    | JWT Token Expiration (7-Day TTL)      | Client must re-login to obtain fresh token; clear 401 Unauthorized message guides user.                  | Expected  |